2 Knowledge Base

2.1 Introduction

The decision to perform an activity internally or obtain it from an external provider is commonly described as a **make-or-buy decision**. Outsourcing represents the transfer of an activity, process, resource, or service from internal organisational control to an external provider. In-house operation, by contrast, retains the activity within the organisational boundary. The choice therefore affects more than the immediate cost of performing an activity. It can also change how resources are controlled, where operational capabilities reside, how dependent the organisation becomes on other parties, and how risks are distributed.

Research on outsourcing has developed across several disciplines, including operations management, strategic management, economics and organisational theory. The literature does not support a simple conclusion that either in-house operation or outsourcing consistently produces better organisational outcomes. Instead, outcomes appear to depend on the characteristics of the activity, the resources and capabilities involved, the transaction environment, supplier relationships, governance arrangements and the organisation's ability to manage the resulting risks. Reviews of outsourcing research have consequently called for more integrated approaches combining operational and strategic considerations (Espino-Rodríguez & Padrón-Robaina 2006; McIvor 2009; Tsay et al. 2018). (Wiley Online Library)

This literature review therefore examines in-house and outsourced operations through two connected perspectives. The first is the **operational perspective**, which concerns performance, control, quality, reliability, capacity, flexibility, continuity and operational dependency. The second is the **strategic/business perspective**, which concerns organisational capabilities, transaction costs, knowledge, strategic dependency, competitiveness and long-term business exposure.

2.2 In-House Operations and Outsourcing

In-house operation allows an organisation to retain direct control over the resources and activities required to perform a process. This may include internal employees, equipment, production capacity, technical knowledge, maintenance resources and management systems. The organisation consequently retains both the benefits and responsibilities associated with the activity.

Outsourcing changes this arrangement by introducing an external organisation into the delivery process. The external provider may supply specialised skills, equipment, capacity, technology or services that the client does not possess internally. Outsourcing can therefore provide access to resources without requiring the client organisation to develop or maintain all of those resources itself.

The literature shows that outsourcing has been used for several reasons, including cost reduction, access to specialised capabilities, increased flexibility, access to technology and concentration on core organisational activities. However, these potential benefits are accompanied by changes in organisational exposure. For example, dependence on an external supplier can introduce supplier performance risk, coordination requirements and reduced direct control. (ScienceDirect)

This means that outsourcing should not be treated simply as a transfer of work. It can be understood as a **change in the organisation's operating structure and risk distribution**.

For this research, the distinction can therefore be represented as:

In-house operation

Organisation → Resources → Process control → Output → Risk retained internally

Outsourced operation

Organisation → Contract/Governance → External provider → Process delivery → Output → Shared/transferred/dependent risk

This distinction provides the basis for comparing the two operating models.

2.3 Transaction Cost Economics

Transaction Cost Economics (TCE) is one of the principal theoretical foundations used to explain make-or-buy decisions. Williamson's work treats the transaction as the basic unit of analysis and examines whether an exchange should be governed internally or through the market. The central concern is to select the governance arrangement that economises transaction costs. (Wiley Online Library)

Transaction costs extend beyond the direct price paid for a service. They may include costs associated with finding suppliers, negotiating contracts, monitoring performance, managing relationships, resolving disputes and adapting agreements when circumstances change. Uncertainty, asset specificity and the frequency of transactions are important considerations within this perspective. (Wiley Online Library)

TCE is particularly relevant to this research because it explains why a seemingly cheaper outsourced activity may introduce additional organisational exposure. If an activity requires highly specific assets, extensive coordination or significant monitoring, the apparent purchase price may not represent its complete economic cost.

The theory therefore provides a basis for examining questions such as:

- How much does the organisation depend on the supplier?
- How difficult would it be to replace the supplier?
- How much monitoring is required?
- How specific are the assets or knowledge involved?
- How uncertain is the operating environment?
- What costs arise when the contractual relationship changes?

TCE consequently provides an important explanation for the **economic and governance dimensions** of operational risk.

However, TCE alone does not fully explain outsourcing decisions. McIvor (2009) argues that transaction-cost and resource-based perspectives need to be considered together because outsourcing decisions also depend on organisational capabilities and the strategic importance of resources. (Wiley Online Library)

2.4 Resource-Based View and Organisational Capability

The Resource-Based View (RBV) considers organisations as collections of resources and capabilities that can contribute to competitive advantage. From this perspective, the outsourcing decision is not only a question of whether an external provider can perform an activity at lower cost. It also concerns whether the activity involves resources or capabilities that are strategically important to the organisation.

Espino-Rodríguez and Padrón-Robaina (2006) identify the RBV as an important complementary perspective to TCE in outsourcing research. Their review distinguishes between explanations concerned with why organisations outsource and those concerned with the relationship between outsourcing and organisational performance. (Wiley Online Library)

The RBV introduces several factors relevant to the present study:

- internal technical expertise;

- organisational knowledge;
- specialised capabilities;
- core activities;
- technological resources;
- learning;
- innovation capability; and
- the ability to develop and maintain capabilities internally.

An organisation may therefore outsource an activity because an external provider possesses a capability that would be expensive or difficult to develop internally. Conversely, an organisation may retain an activity internally when the associated knowledge or capability is strategically important.

This creates a potential trade-off. Outsourcing can provide access to specialised external capabilities while simultaneously reducing the organisation's direct accumulation of internal knowledge. Research on outsourcing and organisational learning has identified knowledge loss, knowledge drainage and reduced organisational learning as potential risks, alongside possible benefits such as access to expertise, quality improvements and greater focus on core competencies. (Swepub)

The RBV therefore contributes primarily to the **strategic/business perspective** of this research.

2.5 Resource Dependence and External Dependency

Outsourcing also changes the dependency structure of an organisation. When an organisation relies on an external provider for an important activity, some control over resources, information or capabilities moves outside the organisational boundary.

Resource Dependence Theory provides a useful perspective for understanding this relationship. Organisations depend on external actors when important resources cannot be completely controlled internally. The resulting dependency can affect organisational behaviour and the balance of power between parties.

Research on outsourcing relationships has applied resource dependence theory to examine how supplier-client relationships are affected by dependency and power. For example, Su, Mao and Jarvenpaa (2014) examined outsourcing relationships from a

resource-dependence perspective and found that responses to external shocks can depend on the power relationship between the parties. (Sage Journals)

For the present research, dependency can therefore be considered in terms of:

- supplier availability;
- number of alternative suppliers;
- supplier substitutability;
- switching difficulty;
- supplier capacity;
- contractual dependence;
- technological dependence;
- access to internal expertise; and
- ability to bring an outsourced activity back in-house.

Dependency is particularly important because an organisation may reduce one form of risk while increasing another. For example, outsourcing may reduce the burden of maintaining internal capacity but increase exposure to supplier disruption.

2.6 Agency Theory and Governance

Outsourcing introduces a relationship in which one organisation contracts another organisation to perform activities on its behalf. This creates an agency relationship in which the client and supplier may have different objectives, information and incentives.

Agency theory is therefore concerned with issues such as information asymmetry, monitoring, incentives, opportunistic behaviour and performance measurement. Reviews of outsourcing theory identify agency theory alongside TCE, RBV and resource dependence theory as an important theoretical explanation of outsourcing decisions. (Sage Journals)

Governance mechanisms are consequently important for managing outsourced operations. These mechanisms can include:

- contracts and service-level agreements;
- performance indicators;
- supplier audits;

- quality inspections;
- performance reviews;
- corrective-action processes;
- supplier risk assessments; and
- regular communication.

The questionnaire developed for this research reflects these mechanisms through its governance section, including contracts, supplier KPIs, supplier meetings, audits, quality inspections, performance scorecards, supplier risk assessments and corrective-action processes.

This provides a direct connection between the theoretical literature and the empirical research instrument.

2.7 Operational Performance

From an operational management perspective, outsourcing can influence cost, quality, delivery reliability, production capacity and flexibility. These variables are particularly important because changes in the operating model can affect the organisation's ability to maintain stable production and service performance.

Research reviewing manufacturing outsourcing has found that the empirical evidence on performance consequences is mixed. Outsourcing has been associated in the literature with outcomes including financial performance, productivity, efficiency, production costs, business performance, market performance and innovation, but findings are not consistently positive or negative. (DOI)

This is important for the present research because it indicates that outsourcing should not be evaluated using a single performance indicator.

For example, an outsourced activity might provide:

- lower direct operating costs;
- greater access to specialised expertise;
- additional capacity;

while simultaneously creating:

- supplier dependency;

- longer response times;
- quality-control challenges;
- reduced direct operational control.

Similarly, an internally performed activity may provide greater control and responsiveness but require higher fixed costs, internal expertise and equipment investment.

The questionnaire therefore measures several operational dimensions rather than treating cost as the sole indicator. The proposed comparison includes cost performance, quality performance, delivery reliability, production capacity and operational flexibility.

2.8 Outsourcing and Operational Risk

Operational risk in an outsourced environment can arise from the interaction between the organisation and its external providers. Supplier failure can affect production, delivery, quality, capacity and continuity.

Lockamy and McCormack (2009) argue that increasing dependence on suppliers can increase an organisation's exposure to supply-network risks. Their work specifically examines supply-network risk as an input into outsourcing decisions. (Taylor & Francis Online)

More recent review evidence also shows that outsourced manufacturing can be exposed to multiple categories of operational risk. A 2026 systematic review of outsourced automotive assembly identified supplier, delay and management risks as major categories, alongside ICT, financial and external-disruption risks. The review emphasised that operational risk emerges from the interaction of supplier capability, organisational dependence, coordination and external conditions rather than from a single supplier characteristic. (Frontiers)

This supports a broader definition of operational risk for the present research.

Instead of defining risk only as the probability of supplier failure, the study considers the consequences of the operating model for:

1. process control;
2. quality;
3. delivery;
4. capacity;

5. flexibility;
6. internal resources;
7. supplier dependency;
8. disruption;
9. response capability; and
10. business continuity.

2.9 Business Continuity and Resilience

Business continuity is particularly important when an outsourced activity is critical to the organisation's operations. If an external provider becomes unavailable, the organisation must have sufficient alternatives to continue the affected activity.

Potential continuity mechanisms include:

- multiple suppliers;
- alternative suppliers;
- safety stock;
- backup production capacity;
- contingency arrangements;
- business continuity plans;
- supplier risk assessments; and
- the ability to bring activities back in-house.

The questionnaire incorporates these mechanisms directly. It asks respondents about preparedness if an important external supplier becomes unavailable and about measures used to reduce disruption, including alternative suppliers, safety stock, backup production capacity and the ability to bring activities back in-house.

This is significant because outsourcing does not necessarily remove risk. It can **change the location and structure of risk**. An internal capacity risk may become an external supplier risk, while an internal fixed-cost exposure may become a contractual or dependency exposure.

Therefore, business continuity provides a bridge between the operational and strategic perspectives.

2.10 Knowledge Retention, Learning and Core Competence

One of the strategic concerns associated with outsourcing is the potential loss of internal knowledge.

Knowledge may be embedded in employees, processes, equipment, routines and organisational experience. When an activity is transferred to an external provider, the organisation may gradually lose some of the practical knowledge previously generated internally.

Research has identified knowledge loss, knowledge drainage and organisational learning problems as potential consequences of outsourcing. At the same time, outsourcing can provide access to external knowledge and specialised expertise. (Swepub)

Hoecht and Trott (2006) additionally identify information leakage as a potential outsourcing risk. Their analysis argues that external specialists may possess access to knowledge from multiple client organisations, creating risks concerning the transfer of firm-specific knowledge and technological competencies. (University of Portsmouth)

Research on outsourcing transitions has also examined the concept of **knowledge at risk**, where the transfer of an activity can create uncertainty about whether critical knowledge and capabilities will remain available to the client organisation. (ScienceDirect)

Consequently, internal knowledge should be treated as both an operational and strategic resource. The present study therefore includes **internal knowledge** and **strategic importance** among the factors considered in make-or-buy decisions.

2.11 Strategic Outsourcing

Strategic outsourcing extends beyond the outsourcing of routine activities. It concerns activities that may influence the organisation's competitive position, capabilities or long-term strategic direction.

McIvor, Humphreys and McIvor's strategic outsourcing research demonstrates the importance of combining transaction-based and capability-based arguments. Their model moves beyond simple cost considerations to consider whether external suppliers can

provide capabilities and value that the organisation cannot achieve as effectively internally. (ScienceDirect)

This perspective is important because an activity can be operationally efficient to outsource while remaining strategically sensitive.

For example, an activity may involve:

- proprietary knowledge;
- critical technical expertise;
- strategic technology;
- important customer relationships;
- intellectual property;
- unique organisational capabilities; or
- knowledge necessary for future innovation.

The strategic decision therefore requires consideration of both current operating performance and future organisational capability.

This supports the use of a dual perspective in the present research rather than treating outsourcing exclusively as an operational efficiency decision.

2.12 Integrated View of Risk

The reviewed literature suggests that the consequences of an operating model can be understood through a sequence of interconnected mechanisms:

Operating Model
↓
Resource Configuration
↓
Control Distribution
↓
Dependency Structure
↓
Operational Exposure
↓
Risk Event

↓

Operational/Business Consequence

↓

Response and Recovery

↓

Residual Risk

Under an in-house model, resources and control remain primarily within the organisation. This can support direct supervision, internal knowledge retention and rapid coordination, but it also means that the organisation retains responsibility for internal capacity, skills, equipment, investment and operating costs.

Under an outsourced model, some resources and execution responsibilities move outside the organisation. This can provide access to specialised resources, capacity and potentially lower operating costs, but may introduce supplier dependency, coordination requirements, information asymmetry and continuity risks.

The literature therefore supports treating **risk as a consequence of the configuration of the operating model**, rather than assuming that one operating model has an inherently higher or lower risk level.

2.13 Theoretical Synthesis

The main theoretical perspectives can be combined as follows:

Perspective	Main question	Relevance to this research
Transaction Cost Economics	Should the activity be governed internally or externally?	Cost, uncertainty, asset specificity, contracting and monitoring
Resource-Based View	What resources and capabilities should remain inside the organisation?	Skills, knowledge, technology, core competence and strategic capability
Resource Dependence Theory	How dependent is the organisation on external resources?	Supplier dependency, substitutability, bargaining position and continuity

Perspective	Main question	Relevance to this research
Agency Theory	How can external performance be controlled?	Contracts, monitoring, incentives, KPIs and governance
Operations Management	How does the operating model affect performance?	Cost, quality, delivery, capacity and flexibility
Knowledge/Capability Perspective	What happens to organisational knowledge and learning?	Knowledge retention, learning, innovation and capability erosion

The literature indicates that no single theory adequately captures the full make-or-buy decision. McIvor (2009) explicitly argues for integrating TCE and RBV, while broader outsourcing literature has combined TCE, RBV, resource dependence and agency perspectives to explain outsourcing decisions. (Wiley Online Library)

This integrated approach is appropriate for the present study because its research questions extend across both operational and strategic/business consequences.

2.14 Research Gap

The literature provides substantial theoretical and empirical discussion of outsourcing, but several limitations remain relevant to this study.

First, much outsourcing research examines specific industries or functions rather than providing a general comparative framework applicable across different industrial activities. Reviews of production and operations management research show that outsourcing has been studied extensively within supply-chain contexts, but the theoretical perspectives and terminology used across strategy, economics and operations management are not always integrated. (Sage Journals)

Second, many studies focus on performance or cost outcomes rather than examining how the operating model changes the **overall risk profile** of the organisation. Manufacturing outsourcing research, for example, reports mixed findings across financial, productivity, cost, business and innovation outcomes. (DOI)

Third, outsourcing risk is multidimensional. Supplier dependency, quality, delivery, capacity, continuity, knowledge, governance and strategic capability can interact. Recent

operational-risk research similarly identifies multiple interacting risk categories rather than a single outsourcing risk factor. (Frontiers)

Fourth, the literature indicates that the effects of outsourcing are context dependent. The same outsourcing arrangement may create different consequences depending on the criticality of the activity, supplier capability, availability of alternatives, internal competence and governance arrangements.

The research gap addressed by this thesis is therefore not simply whether outsourcing is beneficial or harmful. Instead, the study seeks to establish **how the choice between in-house and outsourced operations changes operational and strategic/business risk, which factors explain those changes, and how the resulting risk profile can be represented in a practical comparative framework.**

2.15 Conceptual Framework for the Study

Based on the reviewed literature, the conceptual framework for this research is:

Operating Model
In-house / Outsourced / Mixed
↓
Resource Configuration
Internal resources / External resources / Shared resources
↓
Control Distribution
Direct control / Contractual control / Shared control
↓
Dependency Structure
Internal dependency / Supplier dependency / Multiple-provider dependency
↓
Operational Risk Dimensions
Cost / Quality / Reliability / Delivery / Capacity / Flexibility / Skills / Continuity
↓
Strategic & Business Risk Dimensions
Strategic dependency / Knowledge retention / Core competence / Technology / Financial exposure / Scalability / Bargaining position
↓
Governance & Mitigation
Contracts / KPIs / Audits / Supplier management / Contingency / Backup capacity

↓

Risk Outcome

Reduced / retained / transferred / introduced / amplified / shared exposure

This framework will guide the empirical questionnaire, interviews, data analysis and dashboard development.

The questionnaire already operationalises several elements of this framework. It captures operating model, reasons for internalisation or outsourcing, operational risks, direct performance comparison, business continuity, governance mechanisms and make-or-buy decision factors.

The literature review therefore establishes the theoretical basis for the next stage of the research: converting the conceptual dimensions into measurable variables and testing them using primary evidence.

Core references for the literature review

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This is structured as the **Knowledge Base chapter**, rather than as a generic essay, so it can directly feed the methodology, questionnaire variables, analytical framework and dashboard design.